Comprehensive Analysis Report Template

1) Introduction

This project analyzes the sales dataset to understand overall performance, category and customer behavior, fulfillment patterns, and regional sales contribution. The main goal is to identify where sales are strongest, where variation is high, and which business areas need attention.

2) Data Overview

The data contains the sales- related fields such as Order ID, Amount, Quantity, Category, Size, ship-Country, ship-state, ship-city, Date, fulfilment type. Before enter the analysis, the data should be checked for missing values, duplicates and incorrect data types.

Key dataset attributes:

- Number of records : **128942** x **22**
- Key Variables : [Order ID, Date, Fulfillment, Category, Qty, Amount, ship-state, size, status]
- Data Types : [Categorical, Numerical]

Key Metrices

- Dashboard shows total sales ~ 83.29M and total orders is ~ 128.941K.
- The count of Customers is 120.201K, suggests that the project analyzes a broad customer base with strong commercial scale

3) Missing Data Handling :

Identifying and addressing the missing values is critical to ensure the data realibility.

Key missing data findings :

- Variables with Null values :

  [currency, Amount, ship-postal-code, fulfilled-by]
- Data Treatment :

  Drop the rows for the ship-postal-code, bcz of small percentage of data missing ~0.5%.

  Filling the remaining categorical fields with the mode function.

  And Imputation the numerical fields has moderate, so perform median imputation over the mean.

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4) **Key Findings**
   - Use describe function to analyze the numeric fields and identifies the mean, data dispersion and extreme of sales.
   - To find the top 5 performing states and compare regional sales over the time period.
   - And we perform the correlation, between Quantity and Amount is 0.071 it's indicating a very weak positive relationship. This suggests that Quantity sold has little effect on sales.

5) **C L M**
   - We've to check the customer Lifecycle management over the time period.
   - Customer Acquisition in 2nd month compare to first month is ~ '+26k%', and churn rate in the next two months is ~ '-105%'.
   - The total orders placed by customers in ~'128.94k' over the time.
   - And the sales are weak in last two months, bcz churn rate and little drops in orders count.
6) **Recommendations**
   - If correlation between quantity and amount is strong, pricing and quantity trends are moving together, which supports volume-based strategies. So, focus marketing on the best-performing states and cities, review low-performing categories, and compare fulfillment methods to see which one gives better results.
   - The business should prioritize inventory and promotions around T-shirt, Shirt, Blazer, and Trousers because these categories drive most revenue and volume. Regional campaigns should focus on Maharashtra, Karnataka, and Telangana because these markets contribute the highest sales values.
   - And performs the **CAA** for **New Customer Acquisition** for raises there sales in categories.